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Energy Storages and Technologies for Electric Vehicle

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Abstract: The transport sector is heading for a major changeover with focus on new age, eco-friendly, smart and energy saving vehicles. Electric vehicle (EV) technology is considered a game-changer in the transportation sector as it offers advantages such as eco-friendliness, cheaper fuel cost, lower maintenance expenses, energy-efficient and increased safety. The energy system design is very critical to the performance of the electric vehicle. The first step in the energy storage design is the selection of the appropriate energy storage resources. This article presents the various energy storage technologies and points out their advantages and disadvantages in a simple and elaborate manner. It shows that battery/ultracapacitor hybrid energy system technology is the most suitable for electric vehicle applications. Li-ion battery technology with high specific energy and range is very suitable for this application. Further it analyzes the basic parallel configuration of the hybrid energy storage system.

Keywords- battery, ultracapacitor, electric vehicle, specific power, specific energy

I. INTRODUCTION

Adoption of electric vehicles in the automobile industry is increasing at a rapid pace to reduce dependence on the conventional fossil fuels and for a better pollution free environment. Electric vehicles are zero emission vehicles. The fossil fuel deposits are depleting day by day. In this present scenario, the electric vehicles seem to be the most suitable alternative for a sustainable transportation industry. The carbon dioxide emissions from the transport sector [1] is 32% as shown in Fig.1 which is very significant and if it is properly checked it will have a positive impact on the environment. Similarly the transport sector [1] consumes the majority

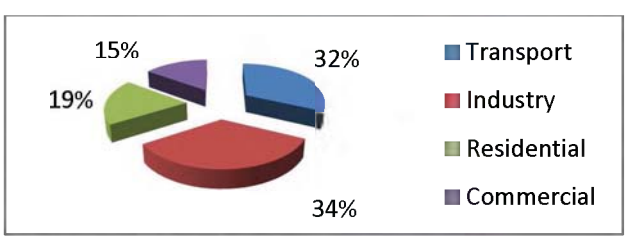

Fig. 1. CO_2emissions of various sectors

of oil produced as shown in Fig.2. Therefore the electric

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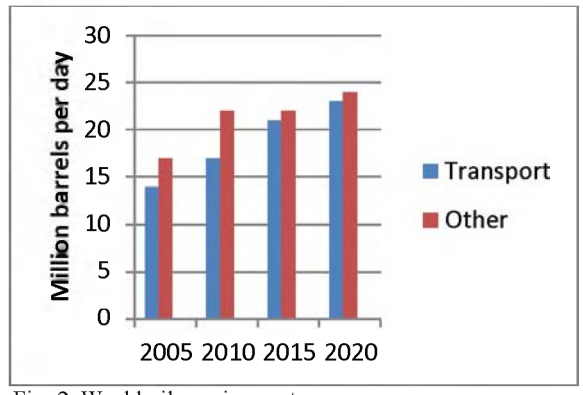

Fig. 2. World oil requirement

vehicle addresses both the problem that is due to on-board energy storage system, it is not dependent on the fuel supply, and at the same time being zero emission vehicle and silent operation, it curbs the noise and air pollution.

II. MOTIVATION

The Government is setting up policies to support the transition to EVs by encouraging financial institutions and commercial banks to provide subsidized interest for financing such vehicles. Recently the state government launched the 'Delhi Electric Vehicle Policy' on August 7, 2020 under which a waiver of registration fee and road tax, and other incentives, and some benefits are being offered (By Money Control News, August 21, 2020). The central government is devising new regulations and policies which are motivating the automobile manufacturers for developing new models for electric bikes, sports utility vehicles (SUVs), electric cars with attractive customer-friendly features and exciting price offers to grab the attention of the consumers are they are inclined for purchasing the same.

Hence the electric vehicles is an emerging field of opportunity in which a lot of research study and work can be done. Cost of the vehicle is the most important deciding factor which motivates the consumer to make the purchase. Electric vehicle have their on-board battery energy storage system which the costliest component of the vehicle. The performance of the vehicle greatly depends on the functionality of this storage system. Hence the storage system selection plays a vital role in the design of the vehicle.

Electric vehicle are not a new invention, they have been

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existing since 1830s. But they have gained importance presently to address the problems of present scenario discussed earlier. Several papers have discussed the methods [2-3] to address the problems related to electric vehicles. The purpose of this article is to give a detailed discussion of the importance of the choice of the on-board storage system in the design of the electric vehicle in a simple and lucid manner.

III. BATTERY-ONLY ELECTRIC VEHICLE

Battery-only electric vehicles have onboard battery energy storage system to meet the power and energy requirements to drive the vehicle. Batteries have inherent characteristics of high energy density and low power density. In real driving conditions there will be transients with peak power demand. With only battery energy storage system, this peak power requirements have to be supplied by the battery pack. Batteries with their intrinsic low specific power are constrained to meet the demand which reduces the driving range of the vehicle. Driving range is measured in the distance travelled by the vehicle till the next charging.

BATTERY TECHNOLOGIES

Several battery system technologies [1,4] with their characteristic values is given in Table 1.

TABLE 1: BATTERY TECHNOLOGIES

Battery type	Specific energy Wh/kg	Specific power W/kg	Battery mass kg for 75km range	Battery mass kg for 150km range
Lead acid	30	250	750	1500
Nickel metal hydride	65	200	346	692
Lithium -ion	90	300	250	500

The values indicated in the table show that Li-ion battery technology with a specific power of 90 Wh/kg is the most suitable option for electric vehicles. At the same time they give a less additional weight to the vehicle for any given range. The weight of the energy storage system adds up to the vehicle mass and has to be included while designing the vehicle tractive power.

IV. BATTERY/ULTRACAPACITOR HYBRID ENERGY STORAGE SYSTEM

Battery/ultracapacitor hybrid energy storage system employ battery as the main energy source and ultracapacitor as the auxiliary source for powering the electric vehicle drivetrain. Ultracapacitors have high specific power [1] of more than 2000 W/kg and low specific energy of 2-5 Wh/kg. Due to this inherent high power density characteristic they can withstand peak power demand requirement during the transients. Batteries cannot withstand deep charge/discharge cycles. But ultracapacitors can withstand the deep discharge cycles and are able to deliver the transient peak power demand.

The need for hybridization is to combine two energy sources so that the advantage of each source is brought out and at the same time the disadvantage of either source is compensated. The working of the hybrid storage system is explained as follows:

1) During peak power demand operations, which may be during acceleration and uphill movement, both the energy sources supply the required power.

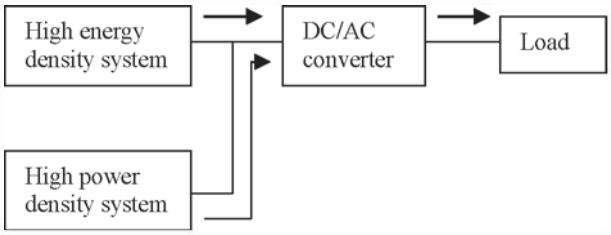

Fig.3: Hybrid energy storage system operation for peak power demand

2) For low power demand operation, which may be during constant speed operation, the high energy density storage will supply the load and the high power density storage system so that it is recharged.

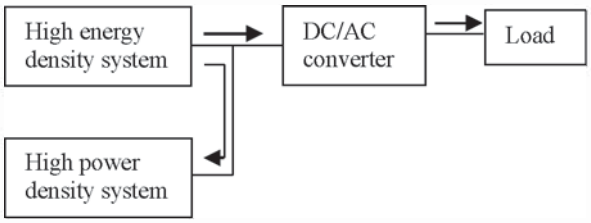

Fig.4: Hybrid energy storage system operation for low power demand

3) During regenerative braking operations, the power will be absorbed by both the energy sources.

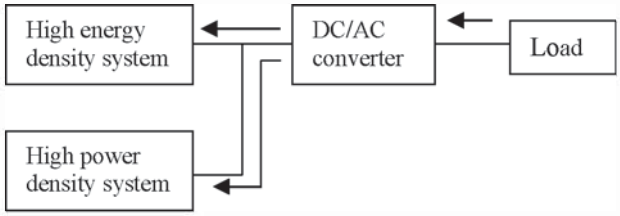

Fig.5: Hybrid energy storage system operation during regenerative braking

The basic configuration during hybridization is to connect the battery unit and ultracapacitor unit in parallel as shown in Fig.6. In this the ultracapacitor reduces the peak battery current and hence the voltage drop across

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the battery unit. But in this configuration the ultracapacitor cannot be used fully due to reduced voltage drop across the battery. Energy storage of ultracapacitor unit is given as:

$$E_{UC}^{rated} = \frac{1}{2} \times C \times V_{rated}^2$$

where E_{UC}^{rated}= rated ultracapacitor energy at the rated voltage V_{rated}. Now suppose the voltage across the battery unit drops by 20%. The remaining ultracapacitor energy is given as:

$$E_{UC}^{rem} = \frac{1}{2} \times C \times (0.8 \times V_{rated})^2$$

The percentage of ultracapacitor energy used is given as:

$$\frac{E_{UC}^{rated} - E_{UC}^{rem}}{E_{UC}^{rated}} \times 100 = 36\%$$

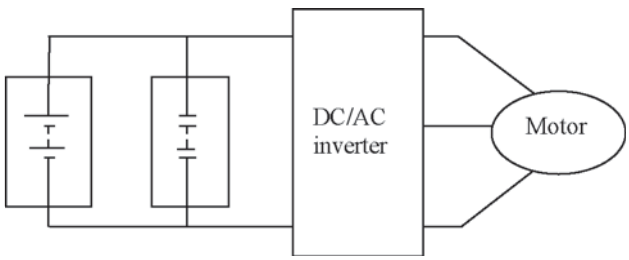

Fig. 6: Basic parallel configuration

Thus only 36% of the ultracapacitor is utilized.

Advantages of battery/ultracapacitor hybrid energy storage system:

- Batteries and ultracapacitors have complementary features and collectively they meet the power and energy demand of the vehicle.
- Battery life span increase as they are not exposed to deep discharge cycles.
- Battery replacement cost is reduced.

Thus battery/ultracapacitor hybrid energy storage system meets the power and energy demand of the vehicle in a cost effective manner.

V. CONCLUSION

This article discusses how the introduction of electric vehicles can address the fossil fuel dependency and increased carbon emissions problems in a appropriate manner. Then it addresses the problem of selecting the storage system for electric vehicle design.

The selection of the energy storage system greatly affects the functionality of the vehicle and is the most important guiding factor in the design of the vehicle. The very purpose of the electric vehicle design is to provide a good driving performance in terms of appreciable range in real driving conditions greatly depends on this selection. Li-ion battery technology is the most suitable option for electric vehicle applications because:

- High specific energy
- Low battery mass for any given range

By far battery/ ultracapacitor hybrid energy storage system is the most viable option available. Both of the sources have complementing features and work coherently so that a good driving performance is experienced. The battery supplies the average power demand due to its low power density feature. The peak power demand during the transients are supplied by the ultracapacitor unit as it can withstand long discharge cycles. The inherent characteristics of the resources are exploited and the problem is addressed effectively. The basic parallel configuration of the battery/ ultracapacitor system is analyzed and it is shown that the ultracapacitor is only 36% utilized due to reduced voltage drop across the battery unit.

Further work can be done as to how to assign the power to the resources so that this selection of battery/ ultracapacitor hybrid energy storage system efficiently gives the performance.

REFERENCES

[1] M. Ehsani, Y. Gao, S. E. Gay, and A. Emadi,"*Modern electric, hybrid electric, and fuel cell vehicles-Fundamentals, theory and design*,"CRC press chp. 1,10.

[2] L. Kouchachvili, W. Yaïci, E. Entchev," Hybrid battery/supercapacitor energy storage system for the electric vehicles", *Journal of Power Sources* 374 (2018), 237–248.

[3] Z. Song, J. Li, J. Hou, H. Hofmann, M. Ouyang, and J. Du, "The battery-supercapacitor hybrid energy storage system in electric vehicle application: A case study," *Elsevier Ltd.,* pp. 433-441, Apr. 2018.

[4] J. Larminie and John Lowry, *Electric Vehicle Technology Explained*, John Wiley & Sons Ltd, chp. 1-3.

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